



Republic of the Philippines
BATANGAS STATE UNIVERSITY
The National Engineering University

Alangilan Campus

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College of Informatics and Computing Sciences

FINAL PROJECT GUIDELINES
CS 324: MODELING AND SIMULATION
Computer Science Department
2nd Semester AY 2025-2026

Final Project Guidelines: Modeling and Simulation
Project Title: Simulation of Traffic Light System

I. General Information

- **Course Title:** Modeling and Simulation (CS 324)
- **Final Project Title:** Simulations of Traffic Light System
- **Type of Project:** Group (10 members per group)
- **Submission Deadline:** May 20, 2026

II. Objectives

- To apply concepts of modeling and simulation in solving real-world problems.
- To understand and simulate the behavior of traffic light systems.
- To develop a working simulation model using any simulation tool/language.
- To analyze system efficiency based on simulation results.

III. Project Description Each group will simulate a **traffic light system** for an intersection. The simulation should:

- Model traffic flow at an intersection controlled by traffic lights.
- Include timing for red, yellow, and green lights.
- Incorporate vehicle queue dynamics and potential congestion.
- Explore different traffic scenarios (e.g., rush hour, low traffic).
- Evaluate how traffic light timing impacts vehicle waiting time and flow.

IV. Suggested Tools/Platforms

- AnyLogic, AnySim, NetLogo, Simul8, MATLAB, Python (SimPy), Java, or other approved tools.
- Use of GUI for visualization is encouraged but optional.

V. Final Project Document Format

1. **Cover Page**
 - University Name
 - Course Title
 - Project Title
 - Group Members (with IDs)
 - Date of Submission
2. **Table of Contents**
3. **Introduction**



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- Background of the project
- Importance of traffic light simulation
- Objectives
- 4. **Literature Review**
 - Brief review of related studies and traffic simulation models
- 5. **Methodology**
 - Description of the model (block diagrams, flowcharts)
 - Parameters used (timing intervals, number of vehicles, etc.)
 - Assumptions made
 - Tools/Software used
- 6. **Simulation Design**
 - Screenshots/illustrations of the simulation
 - Explanation of how the system was modeled
- 7. **Results and Analysis**
 - Present data (tables, charts)
 - Interpret results (vehicle wait time, traffic flow efficiency, etc.)
- 8. **Conclusion and Recommendations**
 - Summary of findings
 - Possible improvements or future work
- 9. **References**
 - Cite books, articles, and tools used
- 10. **Appendices**
 - Source code (if applicable)
 - Additional charts/diagrams

VI. Evaluation Criteria (Rubric)

Criteria	Points
Problem Understanding	10
Simulation Design	20
Functionality and Accuracy	20
Results and Analysis	15
Report Quality	15
Presentation	10
Team Collaboration	10
Total	100



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VII. Reminders

- Maintain academic integrity; plagiarism will result in disqualification.
 - Document each member's contributions.
 - Submit both soft and printed copies if required.
 - Be prepared for Q&A during the presentation.
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For questions, please contact your course facilitator.

Prepared by:

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